Nothing really MATTERS

Zero doesn't always mean *nothing*. If you put a zero on the end of a number, that multiplies it by ten. That's because we use a "place system" in which the *position* of a digit tells you its value. The number 123, for instance, means one lot of a hundred, two lots of ten, and 3 ones. We need zero whenever there are gaps to fill. Otherwise, we wouldn't be able to tell 11 from 101.

A misbehaving *number*

Ask someone this question: "What's 1 × 2 × 3 × 4 × 5 × 6 × 7 × 8 × 9 × 0?" The answer, of course, is zero, but if you don't listen carefully it sounds like an impossibly hard sum. Multiplying by zero is easy, but dividing by zero leads to trouble. If you try it on the calculator built into a computer, the calculator may well tell you off or give you a strange answer like "infinity"!

Dividing equations by zero leads to impossible conclusions. For instance, take this equation:

$$1 \times 0 =$$

If you divide both sides by zero, you get

$$1 = 0 \div 0$$

But if you start with this equation …

$$2 \times 0 =$$

… and do the same thing, you get

$$2 = 0 \div$$

So 1 and 2 equal the same amount, which means that

$$1 = 2$$

And that's impossible. So what went wrong? The answer is that you CAN'T divide by zero, because it doesn't make sense. Think about it – it makes sense to ask "how many times does 2 go into 6", but not to ask "how many times does nothing go into 6".

Happy New Year!

Zero was invented about 1500 years ago, but it's still causing **headaches** even though we've been using it for centuries. When everyone celebrated **New Year's Eve** in 1999, they thought they were celebrating the beginning of a new millennium. But since there hadn't been a year zero, the celebration was a year early. The new millennium and the 21st century actually began on 1 January 2001, not 1 January 2000.